BIS 629: Homework 2

Due on: October 20, 2026 at 3:00 PM on Canvas

This problem is based on MaxART data (Khan et al. 2020*). The MaxART data set "MaxART_HW2.RData" is available in the Homework 2 folder in Canvas. The outcome variable is "Y", where 1 means the viral load at 18 months was $\geq$ 1,000 copies/mL, and 0 otherwise, "treatment" is the treatment indicator (i.e., X in the lecture notes), where 1 indicates that the clinic at enrollment was randomized to early access to ART as soon as testing positive to HIV, and 0 indicates that the clinic at which the participant enrolled was randomized to standard of care , "clinc" is the ID for each clinic in this cluster-randomized trial.

a Estimate the ICC, and the mean and standard deviation of cluster sizes (Hint: You can estimate the ICC from the linear mixed GEE model, using the R package geeglm() with corstr="exchangeable". To use this package properly, you need to sort the data by clinc first).

b Estimate the ESS assuming constant and variable cluster sizes and concisely state the implications for the design of a future study with similar charactersitics.

c. Calculate the Wald test of intervention effect in MaxART with and without considering clustering, using a GEE model with logit link in R. Estimate the intervention effect

* Khan, S., Spiegelman, D., Walsh, F., Mazibuko, S., Pasipamire, M., Chai, B., Reis, R., Mlambo, K., Delva, W., Khumalo, G., Zwane, M., Fleming, Y., Mafara, E., Hettema, A., Lejeune, C., Chao, A., Bärnighausen, T., & Okello, V. (2020). Early access to antiretroviral therapy versus standard of care among HIV-positive participants in Eswatini in the public health sector: the MaxART stepped-wedge randomized controlled trial. Journal of the International AIDS Society, 23(9), e25610.

(95% CI). Create a table to compare results and concisely interpret findings in words.

Sample Code:

```
data <- data[order(data$clinc),]
 #without cluster
fit.nocluster<-geeglm(Y~ treatment, data = data,id=factor(id),
                      family = binomial(link="logit"),
                       corstr="exchangeable")
# with cluster
fit.cluster<-geeglm(Y~treatment, data = data,id=factor(clinc),
                    family = binomial(link="logit"),
+                   corstr="exchangeable")
```

In what follows, we consider the design of a new CRT with features similar to MaxART, as if MaxART were the pilot study for the new study. $\hat{\beta}^*$ from the GEE model considering clustering should be used for the value of $\beta^*_{H_A}$. K_1 can be taken to equal K_2 which can be taken to equal the number of clusters overall divided by two. m_1 can be taken to equal m_2 which can be taken to equal the total sample size divided by the number of clusters divided by two. Later, the variation in m should be assessed to allow for variation in cluster size in the calculations for the new study. If needed, use $\hat{P}(Y = 1|X = 1)$ and $\hat{P}(Y = 1|X = 0)$ as estimated in MaxART.

d. Using you own code, calculate the power for detecting the intervention effect with and without considering clustering under constant and variable assumed cluster sizes, using you own code following formulas discussed in class based upon the Wald test statistic, such as in Lecture 3 part 1 slide 60 and/or part 2 slides 3 and 4. Verify result in PASS. Clearly state all additional input parameters for this power calculation.

e. Using your own code, calculate the minimum number of clusters (i.e., K) needed to detect the difference between p_1 and p_2 with and without considering clustering under

constant and variable assumed cluster sizes, with 80% and 90% power. Let (β_0, β^*) take the value of $(\hat{\beta}_{0|H_A}, \hat{\beta}^*_{H_A})$ from the GEE model considering clustering. Clearly state all additional input parameters needed for this calculate. Verify your results in PASS. Base calculations on the Wald test.

f. Using your own code, calculate the minimum detectable RR given the baseline rate observed in MaxART for 80% and 90% power with and without considering clustering under constant and variable cluster sizes. p_2 can be estimated in MaxART $\hat{\beta}_{0|H_A}$ from GEE model considering clustering, or from a simple 2x2 table. Clearly state all additional input parameters needed for this calculation. Verify your results in PASS. Base calculations on the Wald test.

g. Make a table comparing d-f with and without clustering under constant and variable assumed cluster sizes.

h. Concisely interpret findings from this table in words. Include a footnote providing all background parameters (not varied in the table). How does this compare to the actual study conducted – was it under-powered or over-powered?